

High-Speed In-Kernel Network Periodicity Detection: A Discrete Wavelet Transform Implementation via eXpress Data Path

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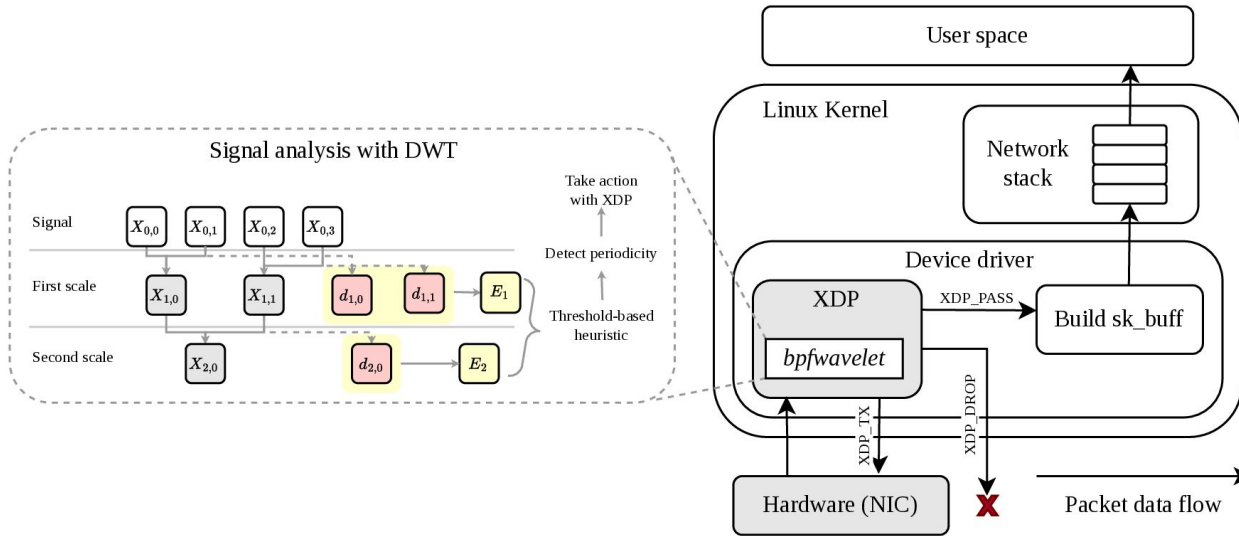


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Introduction & Motivation

- Increasing traffic volume makes it hard to analyse network data, analysis of the streaming data is required.
- Periodicity within the network can indicate malicious or benign activities.
- Discrete Wavelet Transform is an efficient method for analyzing time series data at low computational cost.
- eBPF with XDP allows for computation directly at the Kernel data plane.

Proposed Solution



Challenges

eBPF has limitations:

- No floating point arithmetic support.
- No unbounded loops support.
- Rigorous program checker/validator.

Mathematical Adaptation

The energy equation for the Haar Wavelet: $E_j = \frac{1}{N_j} \sum_k |d_{j,k}|^2, \quad j = 1, 2, \dots, m$

We assume the signal is length a power of two.

This allows us to rewrite the threshold check

$$\frac{E_{j-1}}{E_j} > \frac{\alpha}{\beta}$$

as

$$\beta S_{j-1} > 2\alpha S_j$$

where

$$S_j = \sum_k (X_{j-1,2k} - X_{j-1,2k+1})$$

This way, requiring only integer additions, subtractions, and multiplications.

Streaming Algorithm

- Stores only and accumulator for S and for an intermediary value.
- Requires $2 \times$ decomposition levels + 2 integers. (1)
- Does threshold check. (2)
- Notify operator if detects periodicity. (3)

Algorithm 1 Decompose signal for every new sample

```
1: Initialize  $w, s, c, i \leftarrow 0$  (1)
2: procedure COUNT
3:    $c \leftarrow c + 1$ 
4:   if is new flow then
5:      $\text{Schedule}(\text{Decompose})$ 
6: procedure DECOMPOSE
7:    $x \leftarrow c$ 
8:    $k \leftarrow i$ 
9:   for  $j \leftarrow 0$  to  $L$  do
10:    if  $k$  is even then
11:       $w_j \leftarrow x$ 
12:      break
13:     $s_j \leftarrow s_j + (w_j - x)^2$ 
14:    if  $j \geq 1$  then
15:      if  $\beta \cdot s_{j-1} > 2 \cdot \alpha \cdot s_j$  then (2)
16:         $\text{NotifyPeriodicity}(j)$  (3)
17:     $x \leftarrow x + w_j$ 
18:     $k \leftarrow k >> 1$ 
19:   $i \leftarrow i + 1$ 
20:   $c \leftarrow 0$ 
21:   $\text{Schedule}(\text{Decompose})$ 
```

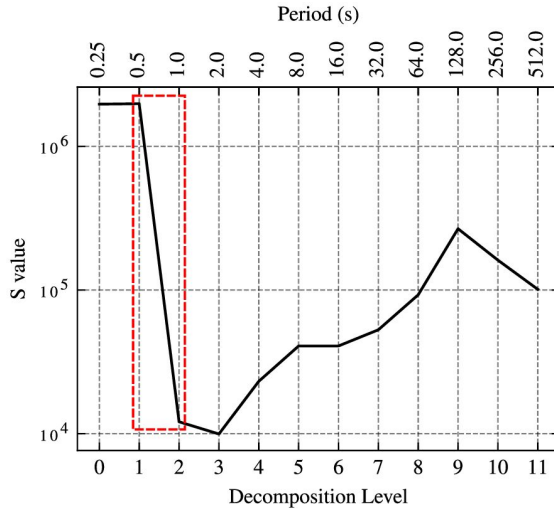
Applicability Evaluation – Setup

We replayed real world, security related, traces with known periodicities through our algorithm and inspected calculated S values.

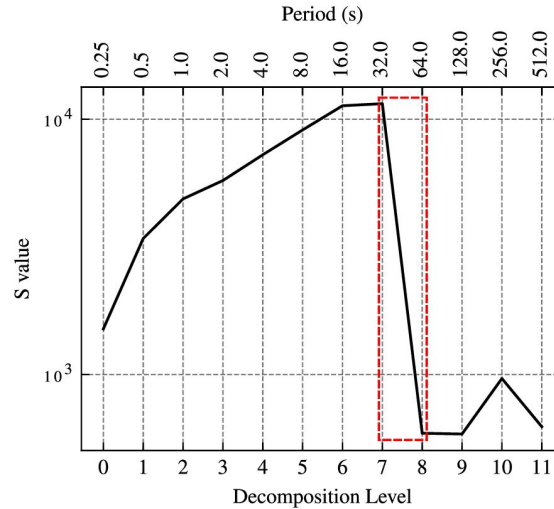
Table 1: Use cases and their periodicities

#	Use Case	Period (s)	Trace Length		Interval (ms)
1	<u>Heartbleed</u>	<u>1</u>	20 min	27 s	250
2	<u>Cobalt Strike</u>	<u>60</u>	17 min	12 s	250
3	Trickbot (A)	300	303 min	47 s	600
4	Trickbot (B)	300	166 min	5 s	600

Applicability Evaluation – Results



Heartbleed

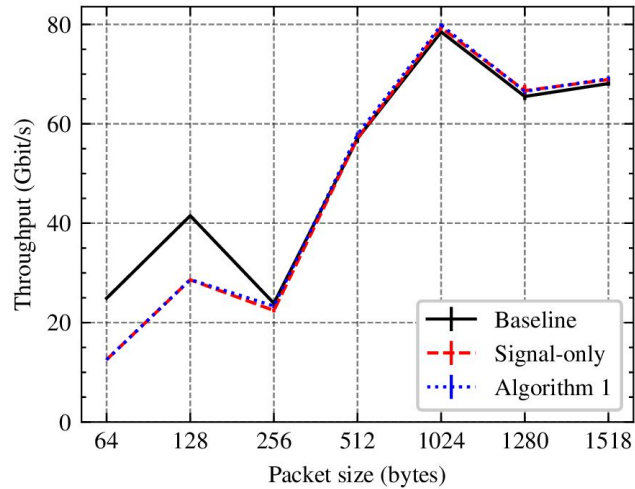


Cobalt Strike

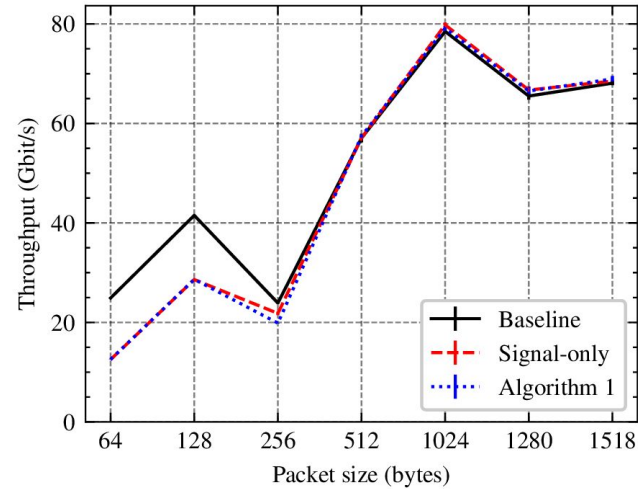
Performance Evaluation – Setup

- Two servers with 100GbE NICs, back-to-back
- Generate packets with Cisco TRex
- xdp-bench as the baseline
- Varied packet sizes, sampling interval, and decomposition levels.

Performance Evaluation – Results (0)

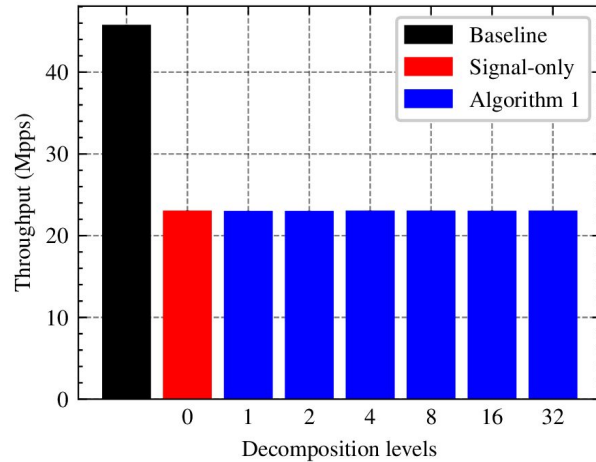


250 μs sampling interval

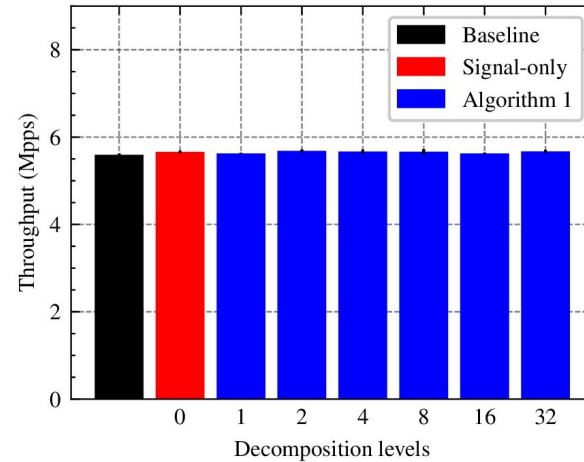


1 s sampling interval

Performance Evaluation – Results (1)



64-byte packets



1518-byte packets

Key takeaways and Future Work

- Signal processing can be used in commodity hardware with Linux Kernel without severe throughput degradation for internet averaged sized packets.
- The number of decomposition levels and the sampling interval had negligible impact on performance, so values can scale for more precision/granularity.
- Future work: use of other Wavelets (known analytical limitations of the Haar wavelet); multiresolution analysis; and integrate with packet-handling capabilities.